

EE671 VLSI Design

Assignment 1

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1 Question 1: Minimum-Size Inverter (INVX1)

1.1 Design Parameters

Table 1: INVX1 Design Parameters

Parameter	Value (μm)
PMOS Width	1.3378974
PMOS Length	0.15
NMOS Width	0.42
NMOS Length	0.15

1.2 Static Characteristics

```
.lib /usr/local/share/pdk/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param L = 0.15
.param Wp = 1.3378974

XM1 vout vin vdd vdd sky130_fd_pr__pfet_01v8
+ l={L}
+ w={Wp}
+ as={Wp*2*L}
+ ad={Wp*2*L}
+ ps={2*(Wp+2*L)}
+ pd={2*(Wp+2*L)}

XM3 n1 vout vdd vdd sky130_fd_pr__pfet_01v8
+ l={L}
+ w={Wp}
+ as={Wp*2*L}
+ ad={Wp*2*L}
+ ps={2*(Wp+2*L)}
+ pd={2*(Wp+2*L)}
```

```

XM2 vout vin 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.42
+ as=0.126
+ ad=0.126
+ ps=1.44
+ pd=1.44

XM4 n1 vout 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.42
+ as=0.126
+ ad=0.126
+ ps=1.44
+ pd=1.44

VDD vdd 0 DC 1.8

Vin vin 0 DC 0
.dc Vin 0 1.8 0.1

.control
run

let gain = deriv(v(vout))

meas dc VM when v(vout)=v(vin)

meas dc VIL when gain=-1 cross=1
meas dc VIH when gain=-1 cross=2

let NML = VIL
let NMH = 1.8 - VIH

print VM
print VIL
print VIH
print NML
print NMH

plot v(vin) v(vout)
.endc
'''

# Results
## Terminal output
'''bash
vm = 9.00001e-01
vil = 7.15014e-01
vih = 1.08533e+00
vm = 9.000008e-01

```

```

vil = 7.150141e-01
vih = 1.085325e+00
nml = 7.150141e-01
nmh = 7.146750e-01
'''
## Voltage transfer curve


```

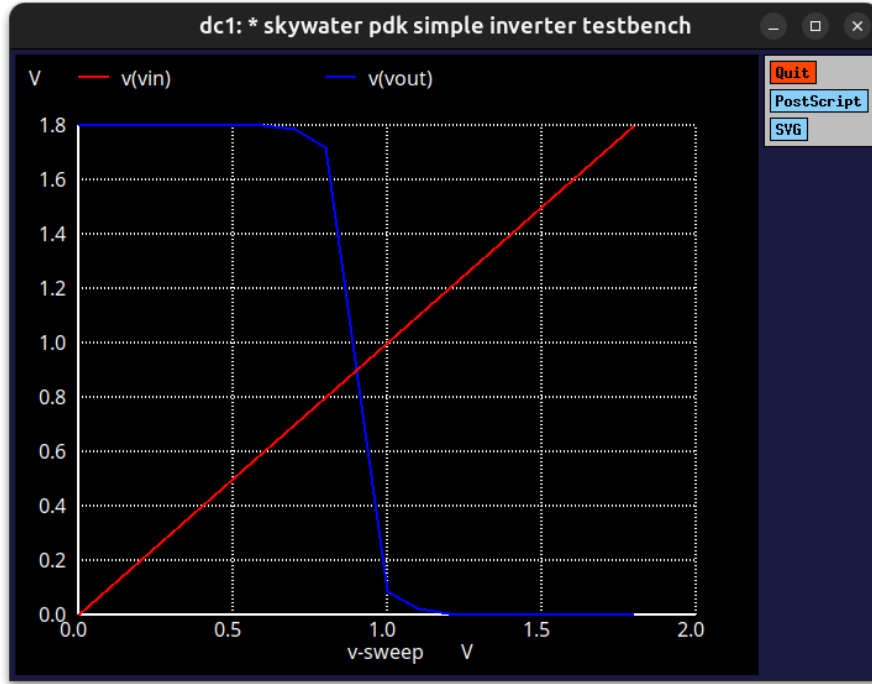


Figure 1: Voltage Transfer Characteristic (VTC) of the INVX1 inverter

Table 2: Inverter Static Characteristics

Characteristic	Value (μm)
$V_{IH}(\text{V})$	1.085
$V_{IL}(\text{mV})$	715.014
$NM_H(\text{mV})$	714.675
$NM_L(\text{mV})$	715.014
Switching Voltage, $V_M(\text{mV})$	900.0008

1.3 Dynamic Characteristics

```
.lib /usr/local/share/pdk/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param L = 0.15
.param Wn = 0.42
.param Wp = 1.3378974
* wp = 3.18547 * wn

XM1 vout vin vdd vdd sky130_fd_pr__pfet_01v8
+ l={L}
+ w={Wp}
+ as={Wp*2*L}
+ ad={Wp*2*L}
+ ps={2*(Wp+2*L)}
+ pd={2*(Wp+2*L)}

XM3 n1 vout vdd vdd sky130_fd_pr__pfet_01v8
+ l={L}
+ w={Wp}
+ as={Wp*2*L}
+ ad={Wp*2*L}
+ ps={2*(Wp+2*L)}
+ pd={2*(Wp+2*L)}

XM2 vout vin 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.42
+ as=0.126
+ ad=0.126
+ ps=1.44
+ pd=1.44

XM4 n1 vout 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.42
+ as=0.126
+ ad=0.126
+ ps=1.44
+ pd=1.44

VDD vdd 0 DC 1.8

Vin vin 0 PULSE(0 1.8 0 20p 20p 500p 1n)

.tran 1p 5n

.measure tran tr
+ TRIG v(vout) VAL=0.36 RISE=1
+ TARG v(vout) VAL=1.44 RISE=1
```

```

.measure tran tf TRIG v(vout) VAL=1.44 FALL=1 TARG v(vout) VAL=0.36 FALL=1
.measure tran tplh TRIG v(vin) VAL=0.9 FALL=1 TARG v(vout) VAL=0.9 RISE=1
.measure tran tphl TRIG v(vin) VAL=0.9 RISE=1 TARG v(vout) VAL=0.9 FALL=1
.measure tran tp PARAM='(tphl+tplh)/2'

.control
run
plot v(vin) v(vout)
.endc
.end

```

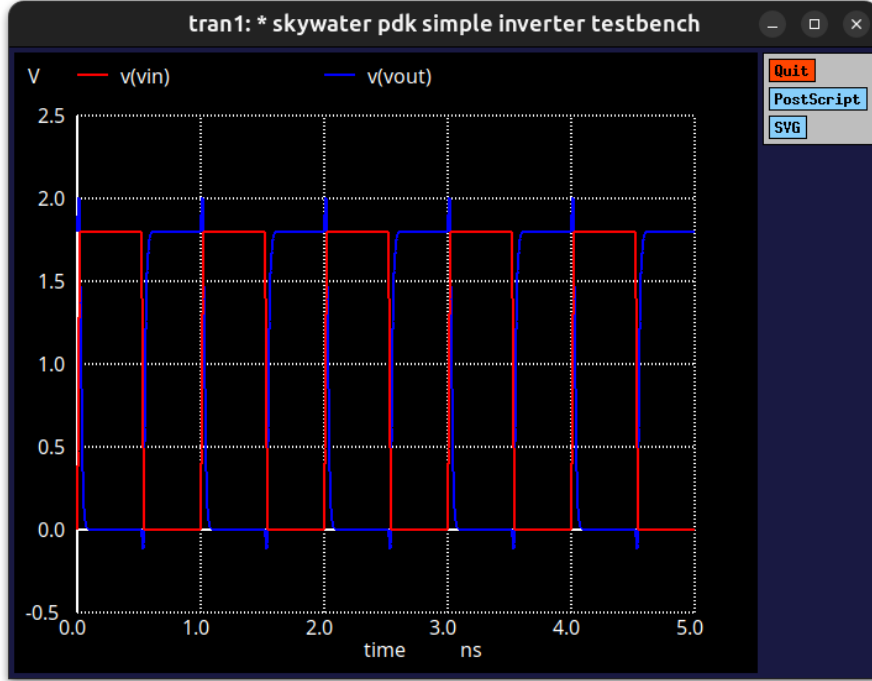


Figure 2: Switching characteristics of the INVX1 inverter

Table 3: INVX1 Dynamic Characteristics

Parameter	Value
Rise Time, t_r (ps)	17.921
Fall Time, t_f (ps)	19.365
Propagation Delay, t_p (ps)	21.414

1.4 Observations and Discussion

1.4.1 Static characteristics

The VTC of the inverter shows a transition around the switching threshold voltage of $V_M \approx 0.9\text{V}$. This shows that the PMOS and NMOS transistors are appropriately sized to become an identical inverter.

The measured values of $V_{IL} = 0.715\text{ V}$ and $V_{IH} = 1.085\text{ V}$ result in noise margins of $NM_L = 0.715\text{ V}$ and $NM_H = 0.715\text{ V}$. The high and low noise margins demonstrate equal noise immunity for both logic states. So we can conclude from these values that the inverter has been properly sized.

1.4.2 Dynamic characteristics

Transient analysis was done using a square input with rise and fall times of 20 ps while driving an identical strength 1 inverter load. The measured rise time and fall time were found to be 17.921 ps and 19.365 ps respectively.

These values of rise and fall times indicates that the selected PMOS width successfully compensates for the lower hole mobility in the PMOS, which leads to nearly symmetrical charging and discharging of the load. The average propagation delay was measured as 21.414 ps.

The small mismatch between the rise and fall times is there because of differences in parasitic capacitances and carrier mobilities. Overall, the inverter achieves balanced switching characteristics while having a low propagation delay.

2 Question 2: Strength-2 Inverter (INVX2)

2.1 Design Parameters

Table 4: INVX2 Design Parameters

Parameter	Value
PMOS Width (μm)	2.6757948
PMOS Length (μm)	0.15
NMOS Width (μm)	0.84
NMOS Length (μm)	0.15

2.2 Static characteristics of INVX2 driving INVX1

```
.lib /usr/local/share/pdk/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param L = 0.15
.param Wn = 0.84
.param Wp = 2.6757948

XM1 vout vin vdd vdd sky130_fd_pr__pfet_01v8
+ l={L}
+ w={Wp}
+ as={Wp*2*L}
+ ad={Wp*2*L}
+ ps={2*(Wp+2*L)}
+ pd={2*(Wp+2*L)}

XM3 n1 vout vdd vdd sky130_fd_pr__pfet_01v8
```

```

+ l=0.15
+ w=1.3378974
+ as=0.40136922
+ ad=0.40136922
+ ps=3.2757948
+ pd=3.2757948

XM2 vout vin 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.84
+ as=0.252
+ ad=0.252
+ ps=2.28
+ pd=2.28

XM4 n1 vout 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.42
+ as=0.126
+ ad=0.126
+ ps=1.44
+ pd=1.44

VDD vdd 0 DC 1.8

Vin vin 0 DC 0
.dc Vin 0 1.8 0.1

.control
run

let gain = deriv(v(vout))
meas dc VM when v(vout)=v(vin)
meas dc VIL when gain=-1 cross=1
meas dc VIH when gain=-1 cross=2

let NML = VIL
let NMH = 1.8 - VIH

print VM
print VIL
print VIH
print NML
print NMH

plot v(vin) v(vout)
.endc

```

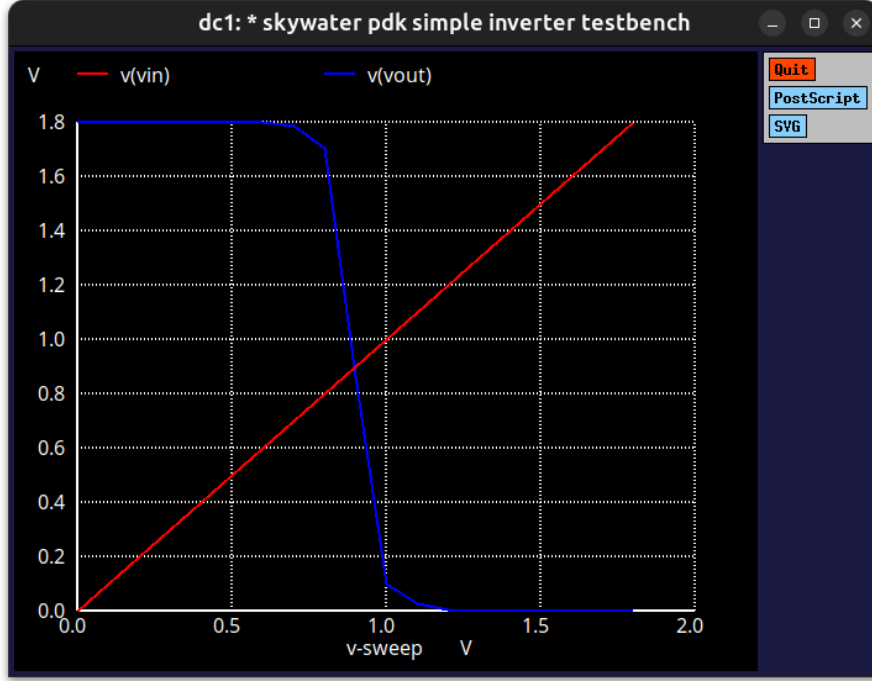


Figure 3: Voltage Transfer Characteristic (VTC) of the INVX2 inverter

Table 5: Inverter Static Characteristics

Characteristic	Value (μm)
$V_{IH}(\text{V})$	1.085
$V_{IL}(\text{mV})$	712.928
$NM_H(\text{mV})$	714.022
$NM_L(\text{mV})$	712.928
Switching Voltage, $V_M(\text{mV})$	896.235

2.3 Dynamic Characteristics

```
* Skywater PDK Simple Inverter Testbench
* AS = AD = W * (2 * L)
* PS = PD = 2 * (W + 2 * L)
.lib /usr/local/share/pdk/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param L = 0.15
.param Wn = 0.84
.param Wp = 2.6757948

XM1 vout vin vdd vdd sky130_fd_pr__pfet_01v8
+ l={L}
+ w={Wp}
+ as={Wp*2*L}
+ ad={Wp*2*L}
+ ps={2*(Wp+2*L)}
```



```

+ pd={2*(Wp+2*L)}

XM3 n1 vout vdd vdd sky130_fd_pr__pfet_01v8
+ l=0.15
+ w=1.3378974
+ as=0.40136922
+ ad=0.40136922
+ ps=3.2757948
+ pd=3.2757948

XM2 vout vin 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.84
+ as=0.252
+ ad=0.252
+ ps=2.28
+ pd=2.28

XM4 n1 vout 0 0 sky130_fd_pr__nfet_01v8
+ l=0.15
+ w=0.42
+ as=0.126
+ ad=0.126
+ ps=1.44
+ pd=1.44

VDD vdd 0 DC 1.8
Vin vin 0 PULSE(0 1.8 0 20p 20p 500p 1n)

.tran 1p 5n

.measure tran tr TRIG v(vout) VAL=0.36 RISE=1 TARG v(vout) VAL=1.44 RISE=1
.measure tran tf TRIG v(vout) VAL=1.44 FALL=1 TARG v(vout) VAL=0.36 FALL=1
.measure tran tplh TRIG v(vin) VAL=0.9 FALL=1 TARG v(vout) VAL=0.9 RISE=1

.measure tran tphl
+ TRIG v(vin) VAL=0.9 RISE=1
+ TARG v(vout) VAL=0.9 FALL=1

.measure tran tp PARAM='(tphl+tplh)/2'

.control
run
plot v(vin) v(vout)
.endc
.end

```

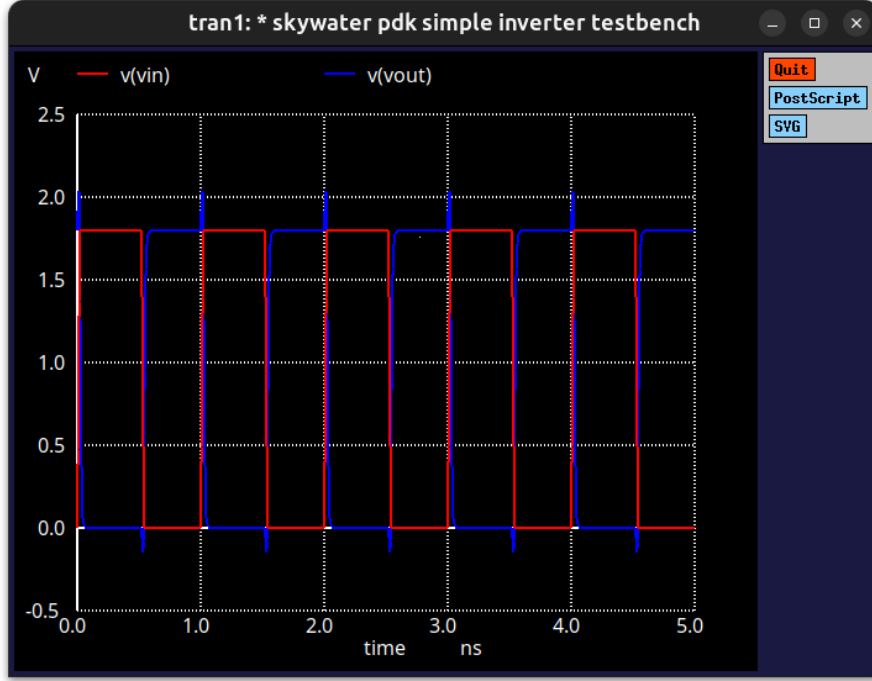


Figure 4: Switching characteristics of INVX2 inverter

Table 6: INVX2 Dynamic Characteristics

Parameter	Value
Rise Time, t_r (ps)	11.534
Fall Time, t_f (ps)	12.046
Propagation Delay, t_p (ps)	16.569

2.4 Comparison with INVX1

Table 7: INVX1 vs INVX2 Comparison

Parameter	INVX1	INVX2
Rise Time (ps)	17.921	11.534
Fall Time (ps)	19.365	12.046
Propagation Delay (ps)	21.414	16.569
V_M (mV)	900.0008	896.235
NM_H (nV)	714.675	714.022
NM_L (mV)	715.014	712.928

2.5 Observations and Discussion

2.5.1 Static characteristics of INVX2 driving INVX1 inverter

The VTC of the INVX2 inverter shows a switching threshold voltage of approximately $V_M = 896.235 \text{ mV}$, which is very close to $V_{DD}/2$. This indicates that scaling both PMOS and NMOS widths by the same factor preserves the symmetry of the inverter.

The measured values of $V_{IL} = 712.928 \text{ mV}$ and $V_{IH} = 1.086 \text{ V}$ result in noise margins of $NM_L = 712.928 \text{ mV}$ and $NM_H = 714.022 \text{ mV}$. These values are nearly identical to those obtained for INVX1, concluding that if we scale the width, then it has very little effect on the static transfer characteristics. The inverter therefore maintains good noise immunity and balanced logic threshold levels while providing higher drive strength.

2.5.2 Dynamic characteristics of INVX2 driving INVX1 inverter

Transient simulations show that the rise time and fall time of the INVX2 inverter are reduced to 11.534 ps and 12.046 ps respectively. Compared to INVX1, both switching times decrease significantly because the larger transistors provide higher charging and discharging currents to the load capacitance.

The average propagation delay decreases from 21.414 ps for INVX1 to 16.569 ps for INVX2. Even though ideally the propagation should ideally reduce by nearly 50% because the widths of both the PMOS and NMOS transistors have been doubled and hence larger transistors can source and sink more current, allowing the output node capacitance to charge and discharge more quickly.

The scaling of the transistors will cause an increase in the capacitances associated with the devices. The increase in the gate area, the source/drain diffusion area, and the source/drain perimeter will lead to an increase in parasitic capacitances such as the gate capacitance, diffusion capacitance, and overlap capacitance. The increase in capacitances means more charges need to be moved during the switching process.

Overall, INVX2 is capable of switching faster with a reduced propagation delay while maintaining almost all static characteristics and noise margins. This shows that transistors can be sized for enhancing inverter drive strength and speed.